

Identifiability and Early-Warning Reliability of Cusp Catastrophe Dynamics in Wearable Electrodermal Recordings

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Abstract—Catastrophe models promise a generative account of overload: functional states as metastable basins of a quartic potential, sudden onset and slow recovery as consequences rather than observations. We ask what such a model can recover from recordings of the length wearable corpora supply, over 12,000 simulated series. Only the relaxation rate and the diffusion coefficient are identified. The parameters carrying the geometry are ratios whose denominator is not bounded away from zero, so length buys their ordering and never their value: rank correlation for the splitting factor reaches 0.88 at 1000 windows while its linear correlation stays at 0.10. That bound holds for any cusp analysis of data of this kind, ours included, and it is the paper’s primary result. We then show that the rolling-window variance and autocorrelation indicators the early-warning literature relies on return exponents of the wrong sign on data simulated from a model that does contain a fold, wrong in 99% of replicates at a 30-sample window, while the exponent computed from the true curvature recovers correctly. The failure is structural: the correlation time diverges at the fold while the window stays fixed. Applying the one prediction that remains identifiable to 147 recordings from 40 people in three open stress corpora returns a surrogate-calibrated null; measured power makes that a bound excluding strong and moderate dynamics in those corpora, not a general absence. The motivating case is overload in autistic adults, for whom no corpus of this density exists; both method results transfer regardless, and a pre-registered validation is specified.

Index Terms—Critical transitions, cusp catastrophe, hysteresis, early-warning signals, identifiability, electrodermal activity, wearable sensing, sensory overload.

I. Introduction

Catastrophe models are an attractive way to formalise overload. A latent load coordinate relaxing in a quartic potential turns functional states into metastable basins and makes abrupt onset with slow recovery a consequence of the geometry rather than an observation to be reported. The motivating case is sensory and executive overload in autistic adults, who report sensory reactivity differences across modalities [1] and in whom sensory atypicality tracks executive difficulty [2]; meta-analysis finds executive dysfunction across the spectrum, with smaller effect sizes in adults [3]. Nothing below depends on that case.

The question this paper answers is what such a model can be estimated to support from a wearable recording, and the answer constrains the approach wherever it is applied.

Existing formal accounts are thin. Computational work on autism is dominated by Bayesian and predictive-coding framings whose empirical support a ten-year review found mixed and methodologically inconsistent [4], and the most recent dynamical treatment of symptom trajectories is avowedly theoretical, collecting no data [5]. Applied state-transition work assigns states from experimental condition labels and counts a transition matrix, which is close to circular: the recovered matrix re-describes the block structure of the protocol and yields nothing observation could contradict.

The cusp formulation avoids that. The load coordinate x_t sits in $V(x; a, b) = \frac{1}{4}x^4 - \frac{1}{2}ax^2 - bx$, where the normal factor b_t carries momentary demand and the splitting factor a_t carries depleted regulatory capacity. The substantive claim is that capacity does not add to load; it changes the geometry. Below a critical a the load tracks demand and returns, above it a second attractor appears, bringing a fold, a hysteresis loop and critical slowing down. Onset and recovery become consequences to check, numerical exponents that data can reject. That is the promise. This paper asks the prior question. We present an identifiability analysis of cusp models fitted to wearable physiology (Sec. IV) and a ground-truth test of the rolling-window early-warning estimators such a model would be checked with (Sec. V). Only two parameters survive a recording of wearable length, and the two carrying the geometry are not among them. That bound belongs to the estimator rather than to our corpora, so it constrains any study of this kind; the estimators, meanwhile, report the wrong sign on data that provably contains a fold. Both results are portable, and both stand independently of the negative empirical result reported here.

Contributions.

- An identifiability bound on cusp models of wearable

physiology (Sec. IV): a 1.2×10^4 -replicate study showing the geometric parameters are ratios with an unbounded denominator, so length recovers their ordering and never their value.

- A ground-truth failure of the standard early-warning estimators (Sec. V): rolling variance and lag-1 autocorrelation report the wrong sign on data containing a genuine fold, with the mechanism identified, plus the eightfold size inflation of the nominal test on near-unit-root series.
- The formulation the bound is stated for (Sec. II): a cusp model whose five states are geometric regions rather than labels and whose 5×5 kernel follows from nine parameters, fitted by exact profiled least squares.
- A calibrated null on 147 recordings, powered so the result excludes strong and moderate dynamics rather than reporting an absence.

Motivation and scope. The programme this work belongs to targets autistic adults, which is why the phenomenology above is the phenomenology of overload. Systematic reviews of wearable sensing [6] and ecological momentary assessment [7] in autism survey what has been collected; none of the corpora they describe is openly released at the temporal density this model needs, and we could not find one elsewhere. None is therefore analysed here, which costs nothing because the results concern estimators. General stress corpora carry the same sensor, sampling rate and recording lengths an autism study would, so a bound measured here transfers to that study directly. No claim about the population transfers, and we make none.

II. The Cusp-Hysteretic Model

A. Fast dynamics and a slow debt variable

The latent load obeys a gradient relaxation with additive noise,

$$dx_t = \lambda[-x_t^3 + a_t x_t + b_t]dt + \sigma dW_t, \quad (1)$$

with control parameters linked to observed drivers and to a slow variable A_t that accumulates recovery debt:

$$b_t = \beta_0 + \beta_S S_t + \beta_T T_t + \beta_U U_t, \quad a_t = \alpha_0 + \alpha_A A_t, \quad (2)$$

$$dA_t = \varepsilon[\text{sp}(x_t - x_{\text{ref}}) - A_t]dt, \quad \varepsilon \ll 1. \quad (3)$$

Here sp is a softplus, S_t , T_t and U_t are sensory, task-switch and uncertainty drivers, and λ sets the timescale without touching the geometry. Load above baseline accrues debt, debt raises a_t , and a higher a_t widens the hysteresis loop, so one episode makes the next easier. That is a mechanistic reading of the refractory period qualitative accounts describe.

B. Folds and two scaling laws

Equilibria solve $x^3 - ax - b = 0$, whose discriminant $4a^3 - 27b^2$ is positive exactly when two stable basins coexist.

Folds sit where $V''(x) = 0$, at $x_{\pm} = \pm\sqrt{a/3}$, giving critical drives $\pm b_c(a)$ with $b_c(a) = \frac{2}{3\sqrt{3}}a^{3/2}$.

Proposition 1 (Hysteresis width, P1). For $a > 0$ the system jumps to the upper branch at $b = +b_c$ and returns at $b = -b_c$, so the loop has width $\Delta b(a) = \frac{4}{3\sqrt{3}}a^{3/2}$.

Linearising (1) about a stable equilibrium gives a relaxation rate $\lambda_{\text{rel}} = 3x^{*2} - a$ that vanishes at a fold, so critical slowing down is a theorem here rather than an assumption. With $\mu = b_c - b$ the distance to the fold, the normal form gives $\lambda_{\text{rel}} \propto \mu^{1/2}$, hence

Proposition 2 (Early-warning exponents, P2). $\text{Var}(x) \propto \mu^{-1/2}$ and $-\log \text{AC1} \propto \mu^{+1/2}$.

These are point hypotheses about exponents, differing in kind from the usual directional claim that an indicator rises before a transition. The sceptical literature notes that such claims hardly ever characterise their error rates, and that they fail when a shift is noise-induced rather than driven by a bifurcation [8], [9].

C. States and transition kernel

Crossing basin membership with the bistability condition partitions the (a, b) plane into five regions (Fig. 1b): calm and focused in the lower basin when monostable, stuck in the lower basin while an overloaded attractor coexists, overloaded in the upper basin, and recovering there once demand has fallen below the level that would have put the system there. The last two matter most: qualitative work uses both words freely, and to our knowledge neither has previously had a precise mathematical meaning. Between-basin moves are noise-induced escapes with Kramers rate $r = \frac{1}{2\pi} \sqrt{|V''(x^*)V''(x_s)|} \exp(-2\Delta V/\sigma^2)$ [10], so the kernel is a deterministic function of (a_t, b_t, σ) : twenty counted probabilities replaced by nine interpretable parameters that move with demand.

III. Materials and Methods

A. Corpora

Three open corpora, taken from their primary repositories rather than mirrors, span a gradient of experimental control: WESAD, a laboratory stress and affect protocol [11]; Wearable Exam Stress, students recorded across examinations [12]; and Nurse Stress, hospital shifts recorded during the COVID-19 outbreak [13]. All supply Empatica E4 electrodermal activity, blood volume pulse, skin temperature and accelerometry. Counted from the released files they give 15, 10 and 15 participants, the nurse corpus covering roughly 1250 h of shifts; windowing and quality screening leave 147 recordings from 40 people, median 194 windows each.

B. Sensor assignment and a negative control

Read the load coordinate and its drivers off the same sensor and the model stops being falsifiable, since shared measurement noise alone manufactures the coupling; we

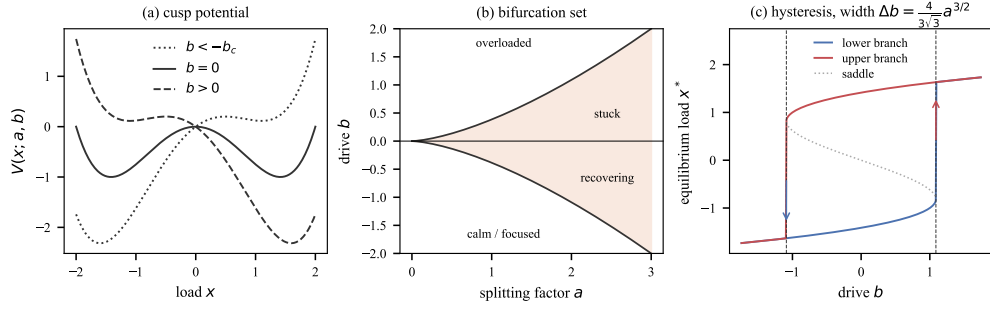


Fig. 1: The geometry from which everything else follows. (a) The cusp potential at three levels of drive b . (b) The bifurcation set in the (a, b) plane, with the five functional states appearing as regions rather than as assigned labels. (c) The hysteresis loop, whose width is fixed by Prop. 1 rather than merely being positive.

therefore assign them to disjoint sensors. Which signal carries the load is settled by measurement, not preference: the coordinate of a slow relaxation process must itself be slow. At a 30s window tonic electrodermal activity has lag-1 autocorrelation 0.98, while a cardiac index from heart rate and RMSSD has 0.19, RMSSD over so few beats being dominated by beat-detection error. Tonic EDA (Configuration B) is therefore the primary channel and skin temperature (C) a robustness check, while the cardiac index (A) is retained as a negative control on which a sound pipeline should find nothing. Both comparisons are internal: the cardiac index serves to justify the sensor assignment and is not offered as a rival method, and each baseline in Sec. V is one of our own nested or competing models.

C. Estimation

We take the Euler map of (1) as the model rather than as an approximation to it. That makes the transition density exactly Gaussian and leaves no discretisation bias to correct, but it is a modelling choice and not a free one: at the λ of interest the two coincide closely, and at larger λ they would not. Expanding the drift, the mean increment is linear in a reparameterised coefficient vector,

$$\Delta x_t = \theta_1(-x_t^3) + \theta_2 x_t + \theta_3 A_t x_t + \theta_4 + \theta_5 S_t + \theta_6 T_t + \theta_7 U_t + \eta_t, \quad (4)$$

with $\theta_1 = \lambda$, $\theta_2 = \lambda \alpha_0$, $\theta_3 = \lambda \alpha_A$ and $\theta_{4:7} = \lambda \beta_{0:U}$. Since A_t depends only on ε , the fit is ordinary least squares profiled over one scalar: exact, no bounds, no starting values, no local optima. We impose $\theta_1 \geq 0$, since a negative value puts the system outside the model rather than merely fitting it badly, and we keep units where that constraint binds so no reported proportion is conditioned on its own outcome.

D. Nulls

The nominal t -test on θ_1 is unusable here. Tonic EDA is close to a unit-root process, and regressing on an integrated series inflates significance in the classical way. We calibrate every test against surrogate ensembles: random-walk surrogates matched on length, increment variance

and marginal scale, and amplitude-adjusted Fourier surrogates (IAAFT [14]) that preserve spectrum and marginal, so only nonlinearity can drive a rejection. Five competing models are scored by held-out one-step predictive log-density: homogeneous Markov, covariate logistic, Gaussian HMM, the nested Ornstein–Uhlenbeck case, and gradient boosting.

E. Settings

Windows are 30s without overlap, swept to 120s as a robustness check. Calibrated tests draw 200 surrogates, 100 in the power and persistence sweeps; IAAFT runs 200 iterations. The Monte Carlo grid uses $n \in \{100, 150, 200, 300, 500, 1000\}$ at 2000 replicates per cell for recovery (1.2×10^4 series), 5000 for size (30,000 nominal, 7,500 calibrated) and 100 for power; the early-warning validation covers 90 configurations (five window widths, two detrending choices, three lengths to $n = 20,000$) at 500 each. Counts were fixed in advance to hold the Monte Carlo standard error on a proportion below 0.01 for size and power, and the AR(1) persistence study runs 1000 replicates at each of nine values of ϕ . All runs derive from one master seed and execute on a single six-core, twelve-thread CPU with ten workers.

IV. What This Data Can Identify

Before testing anything we asked which parameters are recoverable at the series lengths on offer. We simulated with randomised true parameters and refitted, 2000 replicates at each of six lengths, then compared true against estimated values (Table I, Fig. 2a).

Two correlations are reported, because they answer different questions and the gap between them is the finding. Spearman’s ρ asks whether the ordering survives; Pearson’s r asks whether the number can be read off. For λ and σ they agree, and both are high from $n = 100$ up. Everywhere else they diverge.

The splitting factor carries the geometry, so it is the case that matters. Its rank recovery climbs with length, $\rho = 0.33$ at $n = 100$ to 0.88 at $n = 1000$. Its value recovery does not climb at all: $r = 0.15$ at $n = 200$, $r = 0.10$ at $n = 1000$. There $\hat{\alpha}_0$ spans $[-140, 3.3]$ against a 99th percentile

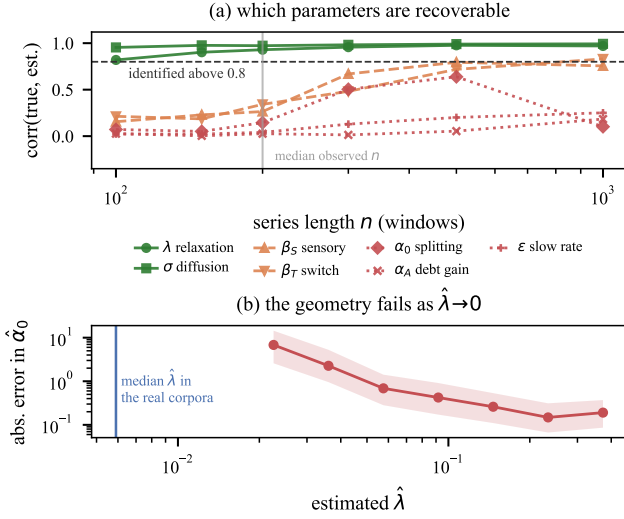


Fig. 2: (a) Correlation between true and estimated values against series length (2000 replicates per point). (b) Error in the recovered splitting factor against $\hat{\lambda}$, growing as $\hat{\lambda}$ falls because $a = \theta_2/\theta_1$; the vertical line is the median $\hat{\lambda}$ of the real corpora.

TABLE I: Recovery of each parameter from simulated series (2000 replicates per cell). Spearman’s ρ measures recovery of the ordering and Pearson’s r recovery of the value. Bold entries mark the two parameters recovered well on both measures.

Parameter	Spearman ρ		Pearson r	
	$n=200$	$n=1000$	$n=200$	$n=1000$
λ (relaxation)	0.94	0.98	0.93	0.97
σ (diffusion)	0.99	1.00	0.97	0.99
β_S (sensory drive)	0.64	0.91	0.26	0.76
β_T (task switch)	0.58	0.88	0.34	0.83
α_0 (splitting)	0.57	0.88	0.15	0.10
α_A (debt gain)	0.32	0.72	0.03	0.18
ε (slow rate)	0.10	0.44	0.04	0.25

of 2.0, so one blow-up in two thousand replicates destroys the linear correlation. This follows from the arithmetic of the model itself and does not reflect any quirk of the simulation.

The consequence applies to any study of this kind. Recovered geometry is built from the ratios $a = \theta_2/\theta_1$ and $b = \theta_4/\theta_1$, and because nothing bounds θ_1 away from zero, both ratios grow without limit as $\hat{\lambda}$ falls (Fig. 2b). This accounts for Table I: division by a small number keeps order, so ρ recovers, but destroys scale, so r does not.

The real fits show the same behaviour. Among units where the constraint does not bind, the median $\hat{\lambda}$ is 0.0059, so even the identifiable parameter, the restoring cubic term, is estimated near zero. The median $\hat{\alpha}_0$ of 12 would place attractors near ± 2 on a unit-variance coordinate; we quote it only to illustrate the ratio pathology, since the preceding paragraph shows it cannot be an estimate, and the substantive reading rests on $\hat{\lambda}$ alone. This median $\hat{\lambda}$ also lies below the whole range of our simulation, placing the recordings deeper in the non-identified regime than

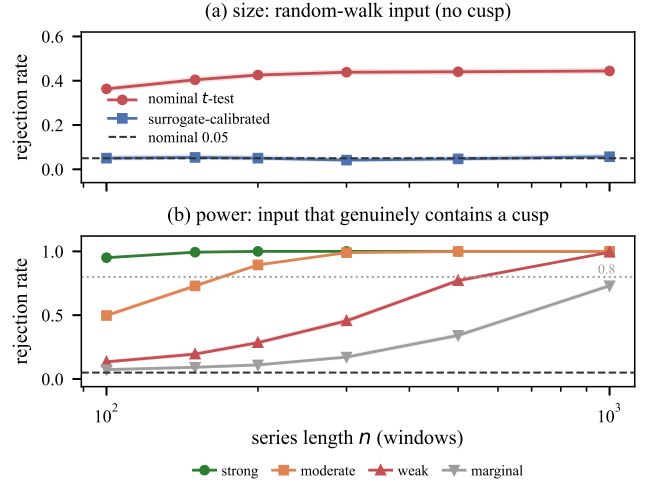


Fig. 3: (a) Random-walk input without a cusp: the nominal t -test rejects 36–44% of the time, while the calibrated test stays near 0.05. (b) Power of the calibrated test for cusp dynamics of four strengths, with Wilson intervals.

any of the 1.2×10^4 synthetic series used to map it. In practice, then, the lesson is narrower than “collect more data”: longer recordings buy a defensible ranking by α_0 but never a reportable value.

Only one of the five pre-stated predictions remains evaluable: the restoring cubic term governed by θ_1 , the sole parameter recovered on both measures. P1 depends on a ratio of two poorly identified quantities, P2 on the distance to the fold, and P3 to P5 on ε and α_A , so we report that structure instead of four numbers presented as tests.

V. Results

A. Validity and power of the significance test

Across 30,000 random-walk replicates spanning six lengths, the nominal one-sided t -test on θ_1 rejects 41.9% of the time instead of 5%. The inflation increases with sample size, from 36.3% (95% CI [35.0, 37.7]) at $n = 100$ to 44.4% ([43.0, 45.8]) at $n = 1000$, as expected from spurious regression on an integrated series and unlike a small-sample artefact. Surrogate calibration restores nominal size on the same grid (5.0% pooled over 7,500 replicates, with every per-length interval covering 0.05). Because calibration costs power, we report power for the calibrated test. At $n = 200$ (2000 replicates per cell) it is 1.00 for strong cusp dynamics, 0.89 for moderate and 0.28 for weak; at $n = 1000$ it is 1.00 for strong and moderate and 0.99 for weak (Fig. 3). The null reported below therefore rules out strong and moderate dynamics in these corpora but cannot exclude a faint fold.

B. The early-warning estimator fails on ground truth

Testing Proposition 2 needs an estimator, and we checked ours before trusting it. On data simulated from a model that genuinely contains a fold, the exponent computed from the true equilibrium curvature comes

TABLE II: Exponent estimates on ground-truth cusp data (1500 replicates per row). The exponent from the true curvature matches theory, whereas the rolling estimators invert at short windows and collapse toward zero at long ones.

Estimator	Pred.	Detr.	Win.	Median	Wrong sign
theory $\lambda_{\text{rel}}(\mu)$	+0.5	—	—	+0.565	—
− logAC1	+0.5	no	30	−0.216	95%
− logAC1	+0.5	no	240	+0.026	32%
variance	−0.5	no	30	+0.307	99%
variance	−0.5	no	240	−0.014	40%

back at +0.565 against a predicted +0.5: theory and implementation agree. The rolling-window indicators the early-warning literature relies on [15], [16] do not. We swept five window lengths, two detrending choices and three series lengths to $n = 20,000$: ninety configurations, 500 replicates each. The failure has two regimes (Table II). At windows of 20 to 120 the sign is systematically inverted, and near-totally so: 99% of replicates carry the wrong sign for variance at window 30, and some configurations invert in all 500. At window 240, both indicators collapse toward zero instead, with medians between -0.018 and $+0.027$, so the sign is effectively a coin flip. Nowhere on the grid does an estimate come near the predicted magnitude.

This is structural, and Sec. II says why. The relaxation rate vanishes as $\lambda_{\text{rel}} \propto \mu^{1/2}$, so the correlation time diverges as $\mu^{-1/2}$ while the window stays fixed: near the fold the window is shorter than the process it measures, which is where Table II shows the inversion. Widening it reaches the other end of the same bias, averaging over the approach itself (the window-240 rows), and detrending removes the low-frequency power carrying what survives. We withdraw our own P2 test as uninterpretable.

The claim is deliberately narrow. It covers fixed-width rolling variance and lag-1 autocorrelation with and without Gaussian detrending, which is what the applied literature mostly runs, and says nothing about adaptive-bandwidth or spectral estimators. It indicts those estimators, not early-warning theory: the exponent from the true curvature recovers on the same data. And it sharpens an existing scepticism [8], [9] by supplying a case where ground truth is known. Within that scope the caution is firm: a directional early-warning result obtained this way is weak evidence either way.

C. The identifiable prediction is not supported

Under the calibrated test the restoring cubic term is significant in 2.7% of the 147 units against a nominal 5%: 0.0% for WESAD ($n = 15$), 0.0% for the examinations ($n = 30$), 3.9% for the nurse sessions ($n = 102$). Since those 102 units come from 15 nurses we also weight one vote per person, giving 1.5% (Wilson $[0.0, 3.1]\%$), with 10% of people showing at least one significant session. That is the most generous reading available, and still what chance produces.

Held-out prediction points the same way. The full model loses to its own nested monostable special case, winning on only 33.8% of units with a median difference of -0.25 nats per step (Wilcoxon $p = 8 \times 10^{-5}$ in favour of the simpler model). Since that special case is the same model with the cubic switched off, this is the clearest evidence available that the term adds nothing. Prospective onset prediction is no better than chance, with AUC between 0.50 and 0.46 at lead times of 3 to 12 windows.

This comparison isolates the cubic term, which is the component the claim depends on. The debt variable and driver set are not ablated in the same way: Sec. IV shows α_A and ε are unidentified at these lengths, so removing them would attribute a change in fit to quantities the data cannot resolve.

D. Structure without fold structure

The null is specific, and calling it noise would be wrong. Two of the three calibrated statistics do exceed their surrogate nulls: the likelihood ratio against the monostable restriction at 40.8% of units, marginal bimodality at 49.7%. Only the statistic isolating the cubic term sits at chance. A bimodal marginal with no cubic drift is exactly the configuration in which distributional evidence alone concludes the opposite of the truth, so it is worth naming what else produces it. Exogenous switching does: a load tracking a driver with two typical levels is bimodal and beats a monostable fit with no restoring cubic anywhere. So does slow non-stationarity, a baseline drifting across a session. Neither is a fold, and separating them from bistability needs the drivers manipulated rather than observed, which is a design requirement and not a longer recording.

Because a test that fires where nothing should be found would undermine the result, we report the control channels explicitly. Configuration B, the primary channel, stays at nominal size under both the random-walk (0.046) and the harder IAAFT (0.069) ensembles across 30s to 120s windows. Configurations A and C reject at 0.11 to 0.16, and A is the negative control.

We tested three explanations and none accounted for this. IAAFT surrogates did not bring C down, which excludes a purely spectral artefact, and clipping rates move opposite to the effect, which excludes saturation. An AR(1) size study across $\phi \in [0, 1]$, with 1000 replicates per point, bounds what persistence alone can produce: the calibrated test stays at or below nominal against random-walk surrogates, peaking at 0.040 at the unit root, and reaches 0.094 against IAAFT. That covers B's IAAFT rate of 0.069 but not A or C, where what remains is nonlinear structure the surrogates do not reproduce.

Two readings survive, and Sec. III separates them on independent grounds. At a 30s window the cardiac coordinate has lag-1 autocorrelation 0.19, too low for the slow coordinate of a relaxation process, so the conservative reading is motion and peak-detection artefact. On A and

C, then, a significant result does not license the word cusp; that is the purpose of a negative control, and it is why the conclusion rests on B alone.

VI. Discussion and Conclusion

Modelling overload as motion in a cusp potential trades a descriptive account for a generative one whose consequences compute in advance: states become basins, the kernel a function of nine parameters, sudden onset and delayed recovery point hypotheses. Precision of that kind can lose, and here it lost twice. The geometry is not recoverable at these lengths, and the one component that is recoverable, the restoring cubic, is neither supported above a calibrated null nor preferred to its own removal on held-out density. The bimodality that survives has readier explanations than a fold (Sec. V).

The identifiability result is the durable contribution, and it is a negative methodological one: cusp geometry recovered from short wearable series is largely an artefact of ratios involving a poorly constrained relaxation rate, and rolling early-warning indicators can invert even when a fold is present. Ranking participants by α_0 needs length, order $n \gtrsim 500$ windows; reporting a value needs a reparameterisation estimating a and b directly, since over the six lengths of Sec. IV length alone never fixed it. A splitting factor of 12 from a 200-window recording is not a finding, and the arithmetic producing it is not specific to our implementation.

Limitations. No autistic participants are analysed, so the motivating claim is untested rather than refuted. Tonic EDA is a peripheral, artefact-prone proxy. Weak cusp dynamics stay compatible with everything here, and the bound is over these corpora. Window width we swept; the softplus debt form, the driver set (S, T, U) and the Euler-map choice we did not. Longer records rescue rankings, not values, and not every parameter even in rank: ε reaches only $\rho = 0.44$ at $n = 1000$.

Toward validation. We pre-specify the study that would test the mechanism in its intended population: 28 days of ecological momentary assessment with continuous wrist physiology in autistic adults and a matched comparison group, using the same estimator, nulls and falsification criteria. The primary hypothesis is a between-group difference in α_0 at $n \gtrsim 500$ windows per participant. P3 to P5 need more than length: with α_A and ε unrecoverable from passive data, they need experimental control over recovery intervals, which would also separate exogenous switching from bistability. A second well-powered null under those conditions would be evidence that overload is not a cusp catastrophe at physiological timescales.

Code and Data Availability

Code, figures and this manuscript are released under the MIT licence at <https://github.com/n-3-0-l-d-3-v/overload-cusp-catastrophe>, with a provenance table recording the replication count behind every number.

The three corpora are third-party and not redistributed: WESAD from the UCI repository (ID 465), Wearable Exam Stress and Nurse Stress from PhysioNet.

Acknowledgment

We thank the authors of the WESAD, Wearable Exam Stress and Nurse Stress datasets for releasing them openly. An AI assistant was used mainly for repeated model test runs, including the simulation and validation runs. The model, the analysis and the final manuscript are the work of the authors, who verified all results and are solely responsible for the content.

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